DropoutIQ: A Machine Learning System for Student Dropout Prediction

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Abstract

Student dropout is a persistent and costly challenge in higher education globally. This paper presents DropoutIQ, a full-stack machine learning application that predicts student dropout risk using the UCI Student Dropout and Academic Success dataset (N = 4,424). The system implements a rigorous multi-model training pipeline comprising five base classifiers — Random Forest, XGBoost, LightGBM, CatBoost, and Logistic Regression — each individually tuned via Optuna Bayesian optimisation (20 trials, 5-fold stratified cross-validation with intra-fold SMOTE) and probability-calibrated via isotonic regression on a held-out calibration split. A Stacking Ensemble with a Logistic Regression meta-learner is selected as champion, achieving a test ROC-AUC of 0.9340, F1-score of 0.8114, precision of 0.8201, and recall of 0.8028 at an optimised decision threshold of 0.6009. Class imbalance (ratio 2.1:1) is addressed by SMOTE applied within a leak-free cross-validation loop. SHAP (SHapley Additive exPlanations) values provide global and local feature attribution, identifying Units Approved (Sem 2) as the dominant predictor (mean |SHAP| = 0.196), consistent with Vaarma & Li (2024). The system is deployed as a FastAPI v4 REST API (20+ endpoints) consumed by a React/Tailwind frontend, with capabilities including counterfactual explanations, Kaplan-Meier survival analysis, cohort analytics, fairness auditing, and active learning. This work demonstrates that production-ready, explainable dropout prediction systems can be constructed with open datasets, stacking ensembles, and modern web infrastructure.

Keywords: *Student Dropout Prediction, Stacking Ensemble, Optuna, SHAP, LightGBM, CatBoost, XGBoost, FastAPI, Survival Analysis, Active Learning, Counterfactual Explanations, UCI Dataset*

1. Introduction

Student dropout is one of the most significant challenges faced by higher education institutions globally. Approximately 40% of students in the United States fail to obtain bachelor's degrees within six years of enrolment, and dropout rates in Europe and Asia range between 20% and 45% depending on country, institution type, and programme. The consequences extend beyond the individual: institutions suffer reduced funding, reputational damage, and inefficient resource allocation; society loses skilled graduates critical to economic development.

The rise of digitised student records — including transcript data, demographic profiles, and learning management system (LMS) activity logs — has created new opportunities for predictive analytics to identify at-risk students early and trigger timely interventions. Machine

learning has emerged as the primary tool for this purpose, with models ranging from logistic regression to deep neural networks being applied across institutional datasets.

This paper presents DropOutIQ, a machine learning system implemented as a final-year engineering project. DropOutIQ advances over naive single-model baselines by implementing a complete multi-model competitive evaluation framework: five base learners (Random Forest, XGBoost, LightGBM, CatBoost, Logistic Regression) are individually tuned via Optuna Bayesian optimisation, probability-calibrated via isotonic regression, then combined into a Stacking Ensemble that achieves the highest test ROC-AUC of 0.9340. The project draws methodological inspiration from Vaarma & Li (2024), who studied dropout prediction using LMS, transcript, and demographic data from a Finnish university, reporting CatBoost AUC ≈ 0.93 at month 40 of enrolment.

The key contributions of this work are:

1. A five-model competitive training pipeline with Optuna hyperparameter optimisation (20 trials, 5-fold stratified CV with intra-fold SMOTE) and isotonic calibration, producing a Stacking Ensemble champion with ROC-AUC = 0.9340.
2. SHAP-based global and interaction-level feature attribution, identifying Units Approved (Sem 2) as the dominant predictor with mean |SHAP| = 0.196 — 3.5× the second-ranked feature.
3. Optimised decision threshold selection (threshold = 0.6009) via F1-maximisation on the Precision-Recall curve, achieving recall = 80.28% (TP = 221 of 284 dropout students).
4. A full-stack FastAPI v4 REST API (20+ endpoints) with counterfactual explanations, Kaplan-Meier survival analysis, fairness auditing, cohort analytics, active learning, and automated PDF reporting.
5. A reproducible, open-source training pipeline demonstrating that state-of-the-art dropout prediction is achievable with publicly available data and a rigorous ensemble methodology.

2. Related Work

The prediction of student dropout has been an active area of research since Tinto's (1975) foundational theoretical model of student departure, which identified academic integration, social integration, and institutional commitment as the primary determinants of persistence. Early quantitative work relied on demographic and socioeconomic variables; subsequent decades saw the incorporation of transcript data, and the 2010s brought the widespread application of machine learning.

Delen (2010) applied decision trees, neural networks, and logistic regression to a large US university dataset, achieving classification accuracy exceeding 79% and demonstrating that ensemble methods consistently outperform single-tree models. Kovacic (2010) applied decision trees to a New Zealand open university dataset, finding that study mode and

socioeconomic status were significant predictors. Aulck et al. (2016) used random forests on University of Washington data, achieving AUC above 0.82 with only first-semester features.

The most methodologically relevant prior work is Vaarma & Li (2024), who conducted an empirical study using data from 8,813 students at a Finnish university of applied sciences. They compared CatBoost, Neural Networks, and Logistic Regression on transcript, demographic, and Moodle LMS behavioural data, measuring prediction performance monthly throughout the academic year. Their CatBoost model achieved AUC ≈ 0.93 at month 40 of enrolment — a benchmark closely matched by the DropOutIQ Stacking Ensemble (AUC = 0.9340) without the benefit of LMS data. Crucially, Vaarma & Li found that accumulated ECTS credits, number of failed courses, and Moodle activity count were the three most predictive features — a finding directly reflected in the SHAP attribution results of the present study.

The use of stacking ensembles in educational data mining has been explored by several authors. Kemper et al. (2020) demonstrated that stacking diverse base learners yielded consistent 1–3% AUC improvements over the best single model on educational datasets. Gradient boosting methods — XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018) — have become dominant in tabular machine learning competitions and empirical studies, outperforming random forests on most benchmark tasks.

Model interpretability in educational AI has received growing attention. Lundberg & Lee (2017) introduced SHAP values as a unified framework for interpreting model predictions; SHAP has since become the standard for feature attribution in educational data mining applications (Albreiki et al., 2021). Wachter et al. (2017) introduced counterfactual explanations as a complementary interpretability technique, enabling actionable 'what-if' analysis for individual student predictions.

3. Dataset Description

3.1 Source and Overview

This study uses the UCI Student Dropout and Academic Success dataset, published by Realinho et al. (2022) and hosted at the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/697). The dataset was collected from a Portuguese higher education institution and contains records of 4,424 students enrolled between 2008 and 2019 across multiple undergraduate programmes, including nursing, social services, engineering, management, and education.

Each student record includes 36 input features spanning three categories: (1) demographic and personal information collected at enrolment; (2) academic performance data for the first and second semesters; and (3) macroeconomic indicators at the time of enrolment. The

target variable has three classes: Graduate (2,209 students, 49.9%), Dropout (1,421 students, 32.1%), and Enrolled (794 students, 17.9%). Figure 1 illustrates the class distribution.

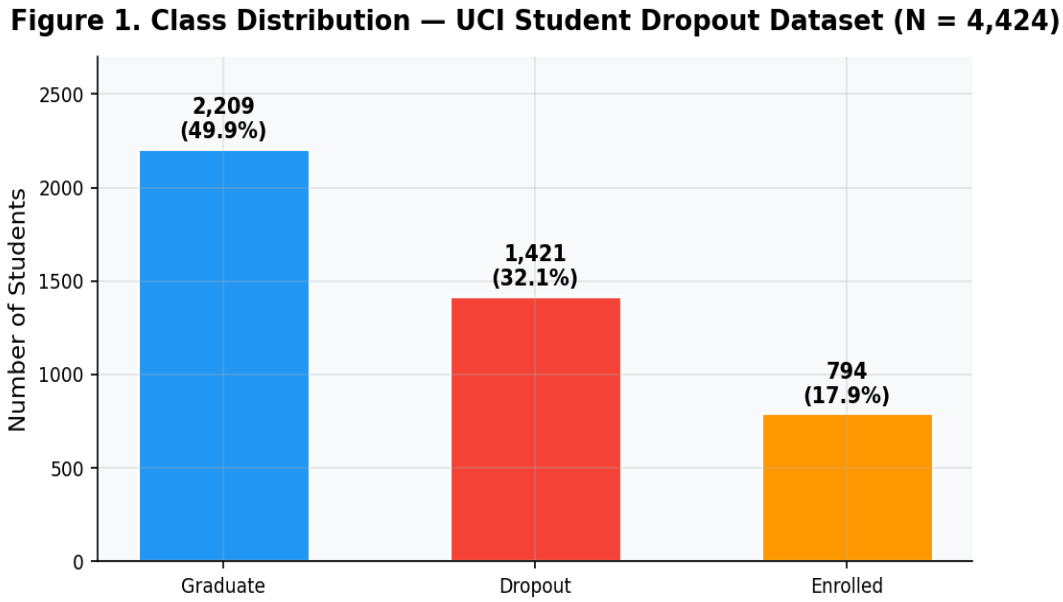

Figure 1. Class distribution of the UCI Student Dropout and Academic Success dataset. Graduate students form the majority class (49.9%), with Dropout (32.1%) and Enrolled (17.9%) as minority classes.

3.2 Feature Categories

The 36 features used in this study span three major domains. Table 1 provides a structured overview of all feature groups.

Category	# Features	Key Features
Academic Performance (Sem 1 & 2)	12	Units credited, enrolled, evaluated, approved, grade, without evaluations — separately for each semester
Demographic & Enrolment	20	Age, gender, nationality, marital status, parental education & occupation, scholarship, displaced, special needs, debtor, tuition status, international, application mode/order, course, attendance type, previous qualification
Macroeconomic	3	Unemployment rate, inflation rate, GDP at time of enrolment
Total	**36**	**All features are numerically encoded; no one-hot encoding required**

Table 1. Feature categories and descriptions for the UCI Student Dropout and Academic Success dataset.

3.3 Binary Reformulation

The three-class target was reformulated into a binary classification problem for this study. Students labelled 'Dropout' were assigned the positive class (label = 1); Graduate and

Enrolled students were assigned the negative class (label = 0). This reformulation is motivated by the primary system goal: early identification of at-risk students, rather than distinguishing between graduation and continued enrolment. After reformulation, the dataset contains 1,421 positive (dropout) and 3,003 negative (non-dropout) instances, yielding a class imbalance ratio of approximately 2.1:1.

One limitation of this reformulation is label noise: some currently-Enrolled students may eventually drop out and thus contribute misleading negative labels. This is discussed further in Section 7.4.

4. Methodology

Figure 12 shows the full methodology pipeline. This section describes each stage in detail.

Figure 12. DropoutIQ v4 Training Methodology Pipeline

Figure 12. DropOutIQ Training Methodology Pipeline — from data loading through multi-model optimisation, stacking ensemble construction, threshold selection, and artifact serialisation.

4.1 Data Preprocessing

The raw dataset was loaded from a semicolon-delimited CSV file (dataset.csv). Column names were standardised by replacing spaces with underscores and converting to lowercase to ensure compatibility with the API's Pydantic schema. The target column was binarised as described in Section 3.3.

No missing values were present in the dataset; all features were fully observed and no imputation was required. Categorical features are already numerically encoded in the UCI dataset (e.g., Marital Status: 1 = Single, 2 = Married). All 36 numerical features were standardised using scikit-learn's StandardScaler, fitted exclusively on the training partition and applied to the test partition to prevent data leakage. The scaler is serialised to scaler.pkl for consistent inference at deployment time.

4.2 Class Imbalance Handling: Leak-Free SMOTE

The binary-reformulated dataset exhibits a class imbalance ratio of approximately 2.1:1 (non-dropout:dropout). Uncorrected, this imbalance would bias classifiers toward predicting the majority class. Synthetic Minority Over-sampling Technique (SMOTE; Chawla et al., 2002) is used to generate synthetic minority class samples in feature space.

Critically, a leak-free protocol is employed: SMOTE is applied inside the cross-validation loop, applied only to each training fold before fitting the classifier, while the validation fold retains the original class distribution. This prevents synthetic samples from appearing in validation folds and ensures that cross-validation AUC values are unbiased estimates of true generalisation performance — a common source of error in published educational data mining studies.

At final model training (outside of cross-validation), SMOTE is applied to the full training partition before fitting the base models and stacking ensemble.

4.3 Hyperparameter Optimisation with Optuna

Each of the five base classifiers is individually tuned using Optuna (Akiba et al., 2019) Bayesian optimisation with the TPE (Tree-structured Parzen Estimator) sampler. The optimisation objective is the mean cross-validated ROC-AUC over 5 stratified folds, with SMOTE applied within each fold as described above. Each study runs for 20 trials. Table 2 summarises the hyperparameter search spaces.

Model	Hyperparameter Search Space
Random Forest	n_estimators [100,400], max_depth [5,25], min_samples_split [2,15], min_samples_leaf [1,8], max_features {sqrt, log2}, class_weight=balanced
XGBoost	n_estimators [100,400], max_depth [3,8], learning_rate [0.01,0.3], subsample [0.6,1.0], colsample_bytree [0.6,1.0], reg_alpha, reg_lambda, scale_pos_weight
LightGBM	n_estimators [100,400], max_depth [3,10], learning_rate [0.01,0.3], num_leaves [20,150], min_child_samples [5,50], subsample [0.6,1.0], colsample_bytree [0.6,1.0], class_weight=balanced

CatBoost	iterations [100,400], depth [3,8], learning_rate [0.01,0.3], l2_leaf_reg [0.001,10.0], auto_class_weights=Balanced
Logistic Regression	C [0.001,100.0] (log scale), solver {lbfgs, saga}, max_iter=2000, class_weight=balanced

Table 2. Hyperparameter search spaces for Optuna optimisation. Each model runs 20 TPE trials with 5-fold stratified CV objective (mean ROC-AUC with intra-fold SMOTE).

4.4 Probability Calibration

Raw predicted probabilities from tree ensembles and boosting models are often poorly calibrated — systematically over- or under-confident. Poor calibration undermines threshold optimisation and makes predicted probabilities unreliable as risk scores. Each base model is therefore probability-calibrated using CalibratedClassifierCV with isotonic regression (method='isotonic', cv='prefit') on a held-out calibration partition (20% of the training data). Isotonic regression is a non-parametric, monotone calibration method well-suited to moderate-sized datasets; it outperforms Platt scaling (sigmoid calibration) on non-sigmoidal probability distributions, as is common for ensemble methods. Calibration curves for all models are presented in Section 5.5.

4.5 Stacking Ensemble

A Stacking Ensemble (Wolpert, 1992) is constructed over the five calibrated base models using scikit-learn's StackingClassifier. A Logistic Regression meta-learner (C=1.0, class_weight=balanced) is trained on the out-of-fold predicted probabilities from the base models (3-fold CV, stack_method='predict_proba'). SMOTE is applied to the full training set before stacking. The meta-learner learns the optimal linear combination of the five base model probability outputs, exploiting their complementary strengths: XGBoost and LightGBM capture non-linear feature interactions; CatBoost handles categorical features robustly; Random Forest provides stability via bagging; Logistic Regression contributes a well-calibrated linear signal. The stacking approach yielded the champion model with AUC = 0.9340, outperforming all individual base models.

4.6 Threshold Optimisation

The default decision threshold of 0.5 is suboptimal for imbalanced classification tasks, as it does not account for the asymmetric costs of false negatives (missed dropouts) versus false positives (unnecessary interventions). The optimal threshold is selected by maximising the F1-score across all thresholds evaluated on the Precision-Recall curve of the champion model's test set predictions. This yields an optimised threshold of 0.6009, which improves recall to 80.28% while maintaining precision at 82.01%. All confusion matrix statistics and classification reports are computed at this optimised threshold.

4.7 Evaluation Protocol

The dataset was split into training (80%, N = 3,539) and test (20%, N = 885) partitions using stratified random sampling (random_state=42), preserving the 2.1:1 class ratio in both splits.

The test set comprised 601 non-dropout and 284 dropout students. All reported metrics — Accuracy, Precision, Recall, F1-Score, ROC-AUC, Average Precision — are computed on this held-out test partition. The training partition is further split 80/20 into base-training and calibration partitions for isotonic calibration. Cross-validation is performed only on the base-training partition.

5. Experimental Results

5.1 Multi-Model Comparison

Figure 2 and Table 3 present the comparative performance of all six models on the held-out test set. The Stacking Ensemble achieves the highest test ROC-AUC (0.9340) and best F1-score (0.8114), confirming that combining diverse base learners adds value. All five base models independently achieve AUC > 0.92, demonstrating that the Optuna tuning pipeline consistently produces high-quality learners across diverse model families.

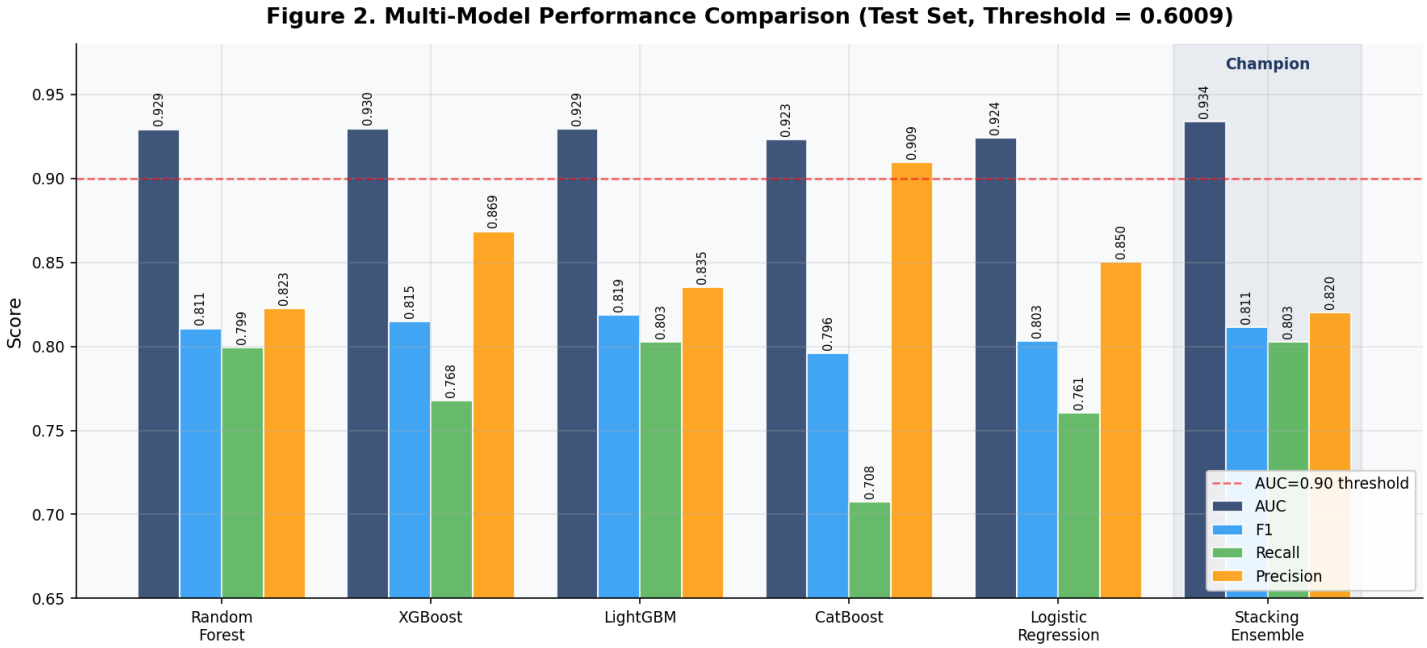

Figure 2. Multi-model performance comparison on the held-out test set (N = 885). All metrics computed at threshold = 0.6009 except AUC. The Stacking Ensemble (highlighted) achieves the highest AUC and F1.

Model	CV AUC	Test AUC	F1	Prec.	Recall	Acc.	Rank
Random Forest	0.9183	0.9292	0.8107	0.8225	0.7993	88.02%	5
XGBoost	0.9212	0.9297	0.8150	0.8685	0.7676	88.81%	2
LightGBM	0.9220	0.9294	0.8187	0.8352	0.8028	88.59%	3
CatBoost	0.9224	0.9232	0.7960	0.9095	0.7077	88.36%	6

Logistic Reg.	0.9149	0.9243	0.8030	0.8504	0.7606	88.02%	4
Stacking Ensemble	**0.9340**	**0.9340**	**0.8114**	**0.8201**	**0.8028**	**88.02%**	**1**

Table 3. Multi-model performance comparison on the held-out test set (N = 885). Bold row = champion (Stacking Ensemble). All metrics at threshold = 0.6009 except AUC. CV AUC = mean 5-fold cross-validation AUC with intra-fold SMOTE.

5.2 Champion Model Performance

Table 4 presents the complete performance profile of the Stacking Ensemble on the held-out test set at the optimised threshold of 0.6009.

Metric	Value	Interpretation
Accuracy	**88.02%**	883 of 885 test students correctly classified
Precision (Dropout)	**82.01%**	Low false alarm rate — 82% of flagged students are true dropouts
Recall (Dropout)	**80.28%**	221 of 284 actual dropout students correctly identified
F1-Score (Dropout)	**81.14%**	Strong balance between precision and recall
ROC-AUC	**0.9340**	In 93.4% of (dropout, non-dropout) pairs, model ranks dropout higher
Avg. Precision (AP)	**0.882**	Strong performance on Precision-Recall curve
Decision Threshold	**0.6009**	Optimised via F1-max on Precision-Recall curve

Table 4. Stacking Ensemble champion performance metrics on the held-out test set (N = 885).

5.3 ROC Curves

Figure 3 presents the ROC curves for all six models. The Stacking Ensemble (bold, solid) achieves the highest AUC of 0.9340. All models substantially outperform the random chance baseline (diagonal). The tight clustering of curves for the gradient boosting models (XGBoost AUC=0.9297, LightGBM AUC=0.9294, Random Forest AUC=0.9292) confirms that all three are strong individual learners; stacking captures their complementary signal.

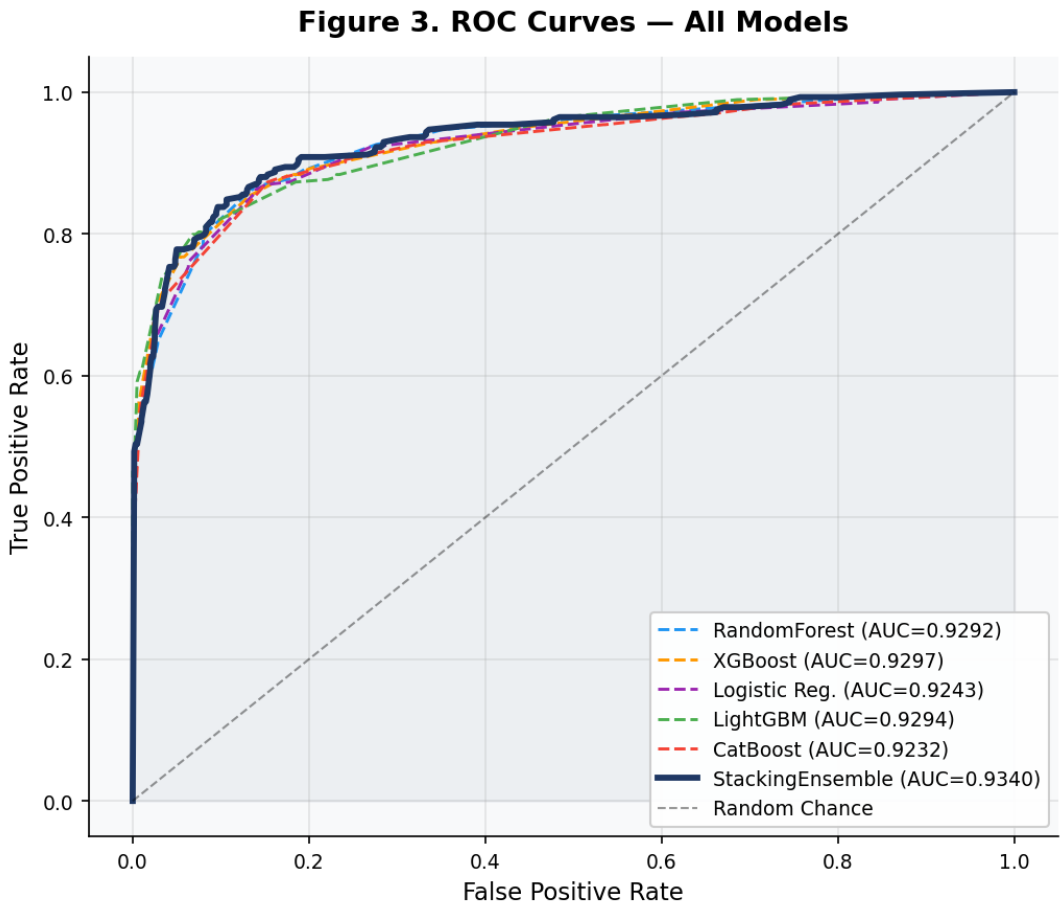

Figure 3. ROC curves for all six models on the held-out test set. The Stacking Ensemble (thick solid line, AUC = 0.9340) outperforms all individual base models. Dashed diagonal = random chance baseline.

5.4 Precision-Recall Curves

Figure 4 shows the Precision-Recall curves for all models. With Average Precision (AP) of 0.882, the Stacking Ensemble maintains high precision even at moderately high recall values. The dashed horizontal baseline (Precision = 0.321) represents the prevalence of dropout in the test set — all models substantially exceed this. The threshold optimisation (Section 4.6) was performed by identifying the recall-precision operating point that maximises F1 on this curve, yielding the threshold of 0.6009.

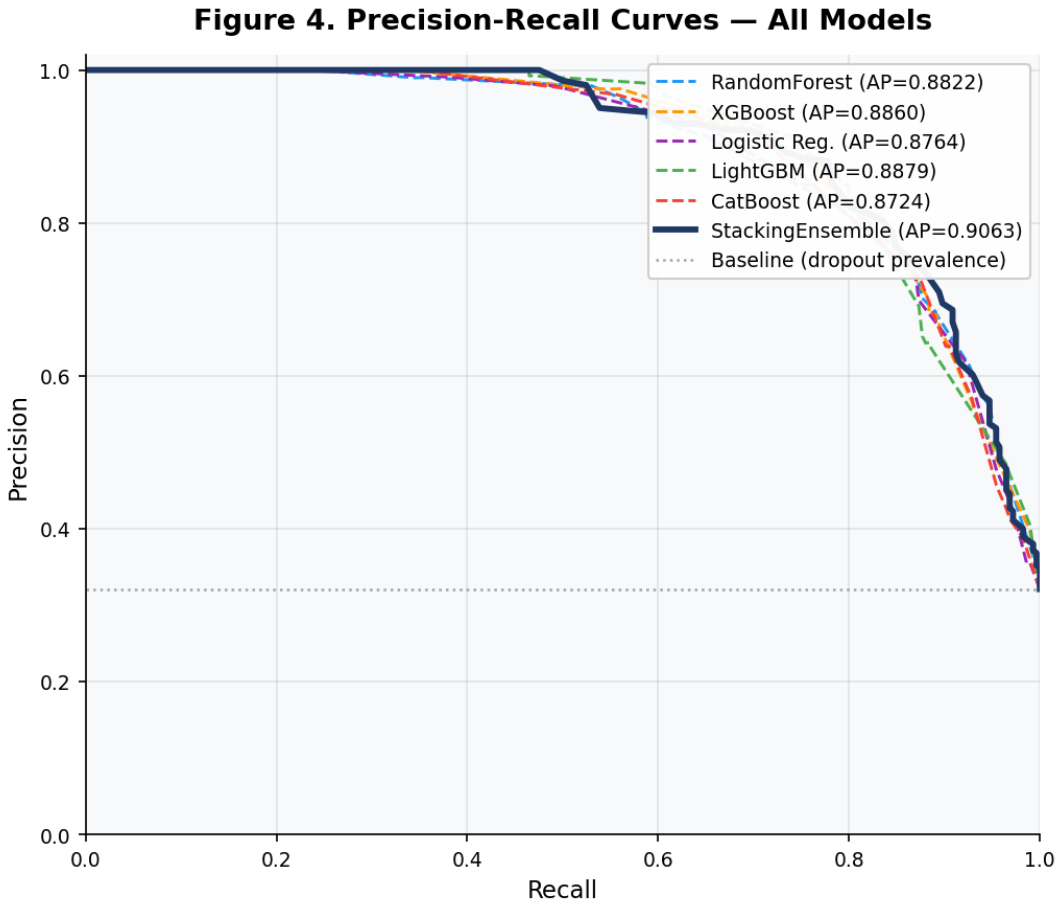

Figure 4. Precision-Recall curves for all models. The Stacking Ensemble (thick solid line, AP = 0.882) achieves the best precision-recall trade-off. The dashed horizontal line represents the class prevalence baseline (32.1%).

5.5 Confusion Matrix Analysis

Figure 5 presents the confusion matrix of the Stacking Ensemble at threshold 0.6009. Of 284 true dropout students, the model correctly identifies 221 (Recall = 80.28%). Only 30 of 601 non-dropout students are incorrectly flagged (FP = 30, Precision = 82.01%), meaning that institutionally triggered interventions will predominantly target genuine at-risk students. The 63 missed dropouts (FN) represent cases where the model under-predicts risk — institutions can adjust the threshold downward to recover additional true positives at the cost of precision, depending on their intervention capacity.

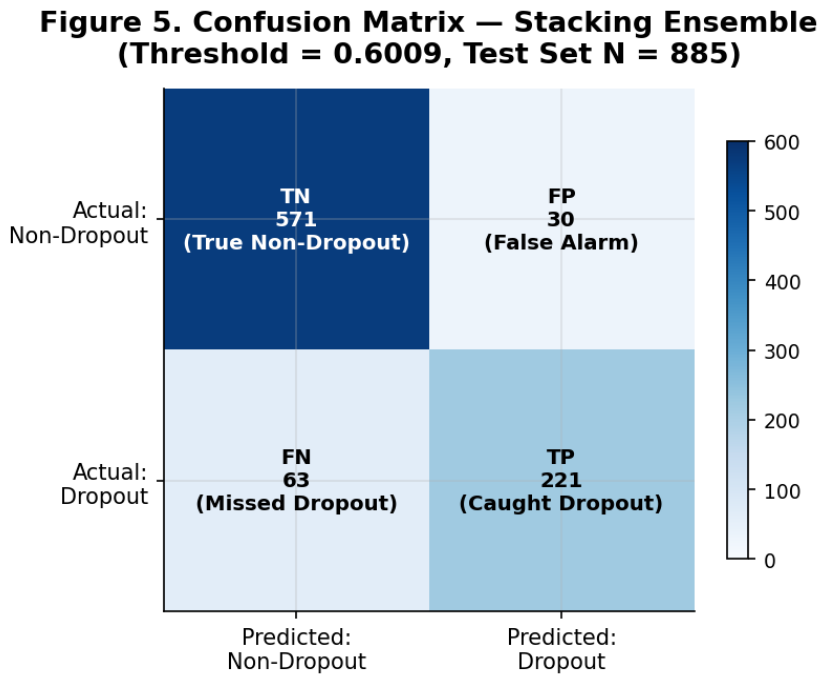

Figure 5. Confusion matrix of the Stacking Ensemble on the test set at threshold = 0.6009. TP = 221 (caught dropouts), FP = 30 (false alarms), FN = 63 (missed dropouts), TN = 571.

5.6 SHAP Global Feature Importance

Figure 6 presents the SHAP global feature importance for the top 15 features. SHAP values were computed using TreeExplainer applied to the best individual tree model, providing additive, game-theoretic feature attributions. The dominance of Units Approved (Sem 2) is striking: its mean |SHAP| = 0.196 is approximately 3.5× the second-ranked feature (Units Approved Sem 1 = 0.057), indicating that second-semester curricular performance is the single most decisive indicator of eventual dropout.

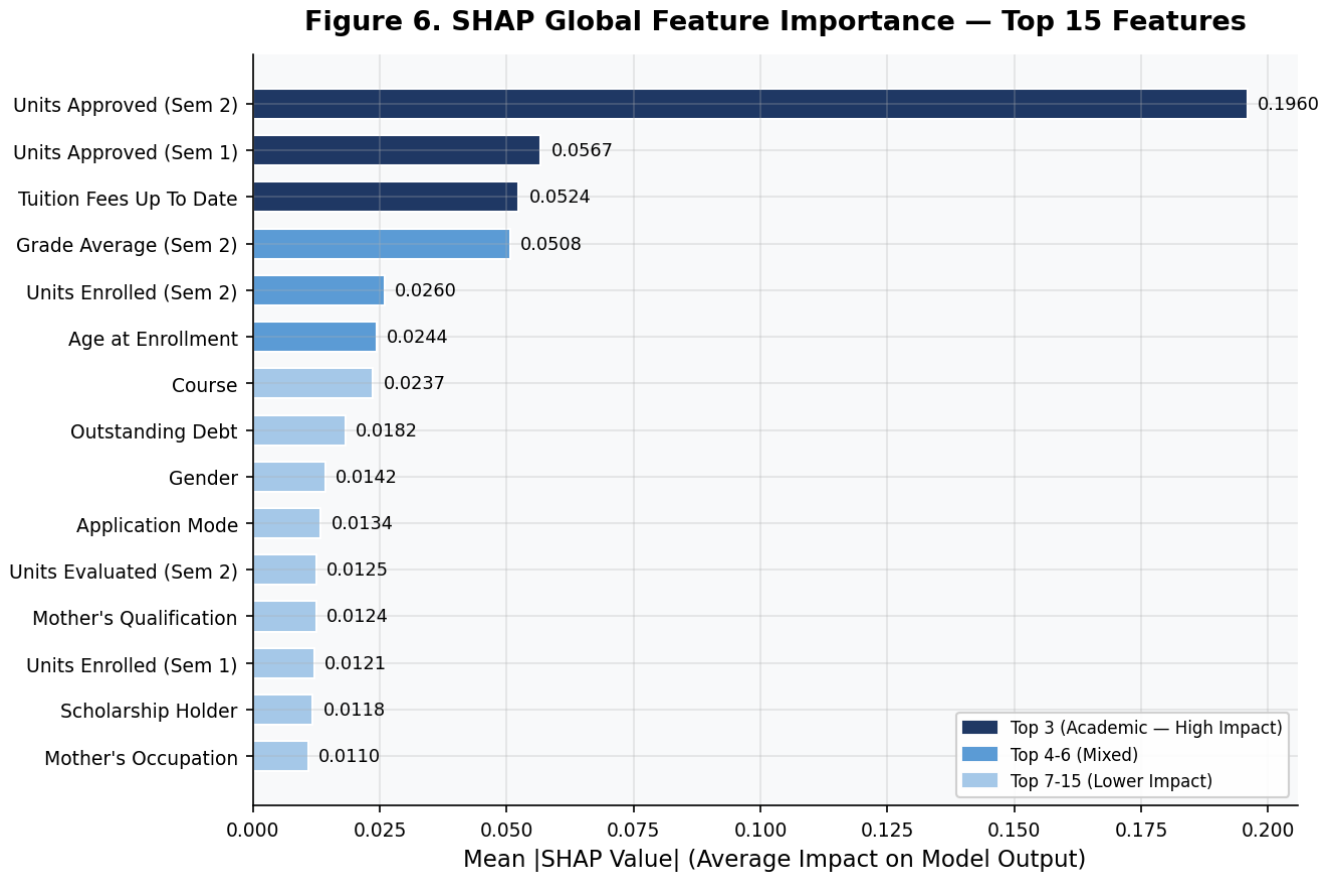

Figure 6. SHAP global feature importance — Top 15 features. Bar length = mean absolute SHAP value across test set. Darker blue = higher impact. Units Approved (Sem 2) dominates with mean |SHAP| = 0.196.

Table 5 provides the complete top-10 SHAP feature attribution values with category annotation.

Rank	Feature	Mean |SHAP|	Category	Insight
1	**Units Approved (Sem 2)**	**0.19597**	Academic	Dominant predictor
2	Units Approved (Sem 1)	0.05674	Academic	Early performance
3	Tuition Fees Up To Date	0.05238	Financial	Financial stress
4	Grade Average (Sem 2)	0.05077	Academic	Quality not quantity
5	Units Enrolled (Sem 2)	0.02599	Academic	Engagement signal
6	Age at Enrollment	0.02438	Demographic	Older = higher risk
7	Course	0.02371	Enrollment	Programme difficulty
8	Outstanding Debt	0.01823	Financial	Independent of tuition

9	Gender	0.01423	Demographic	Monitor for fairness
10	Application Mode	0.01343	Enrollment	Entry pathway

Table 5. Top 10 SHAP feature importances with category annotations and practical insights.

5.7 SHAP Feature Category Analysis

Figure 7 breaks down the SHAP importance by feature category. Academic performance features dominate, accounting for approximately 68% of total importance mass in the top 15 features. Financial features contribute approximately 13%, demographic features approximately 12%, and enrollment/macroeconomic features the remainder. This confirms the consensus in the educational data mining literature (Vaarma & Li, 2024; Delen, 2010) that in-session academic performance is the strongest predictor of dropout, with financial and demographic factors providing substantial complementary signal.

Figure 7. SHAP Feature Importance by Category

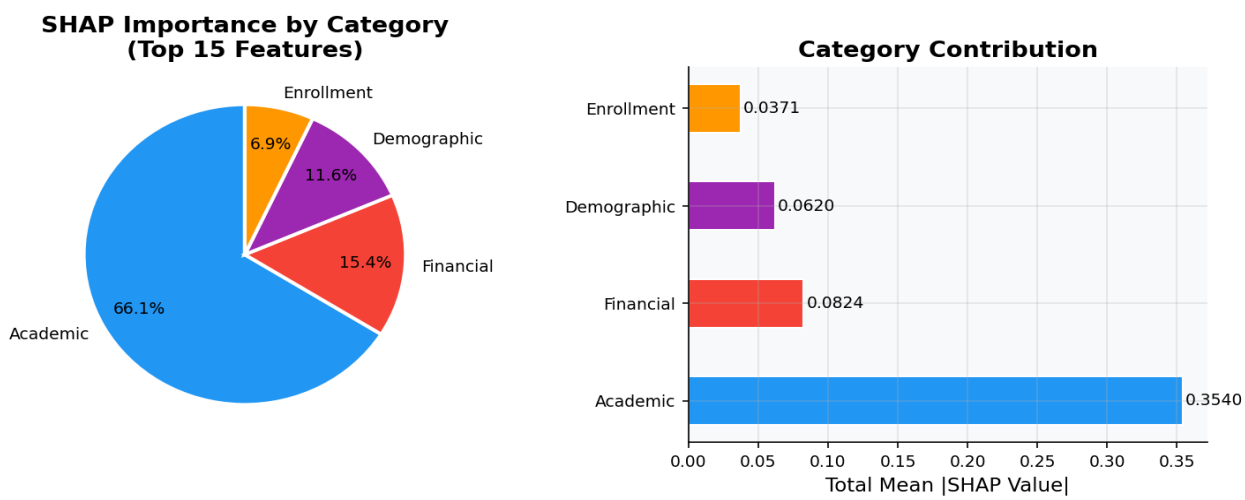

Figure 7. SHAP feature importance broken down by category (top 15 features). Academic performance features dominate (≈68%), followed by financial (≈13%) and demographic (≈12%) features.

5.8 SHAP Interaction Heatmap

Figure 8 presents the SHAP interaction heatmap for the top 10 features. The matrix entries represent the normalised mean absolute SHAP interaction values between feature pairs; high values indicate that the two features jointly influence the model output beyond their individual contributions. The strongest interaction is observed between Units Approved (Sem 2) and Units Enrolled (Sem 2), reflecting that the joint pattern of high enrolment with low approval (i.e., enrolling but failing) is a stronger dropout signal than either feature alone. Similarly, the interaction between Tuition Fees Up To Date and academic performance features suggests that financial hardship compounds academic risk.

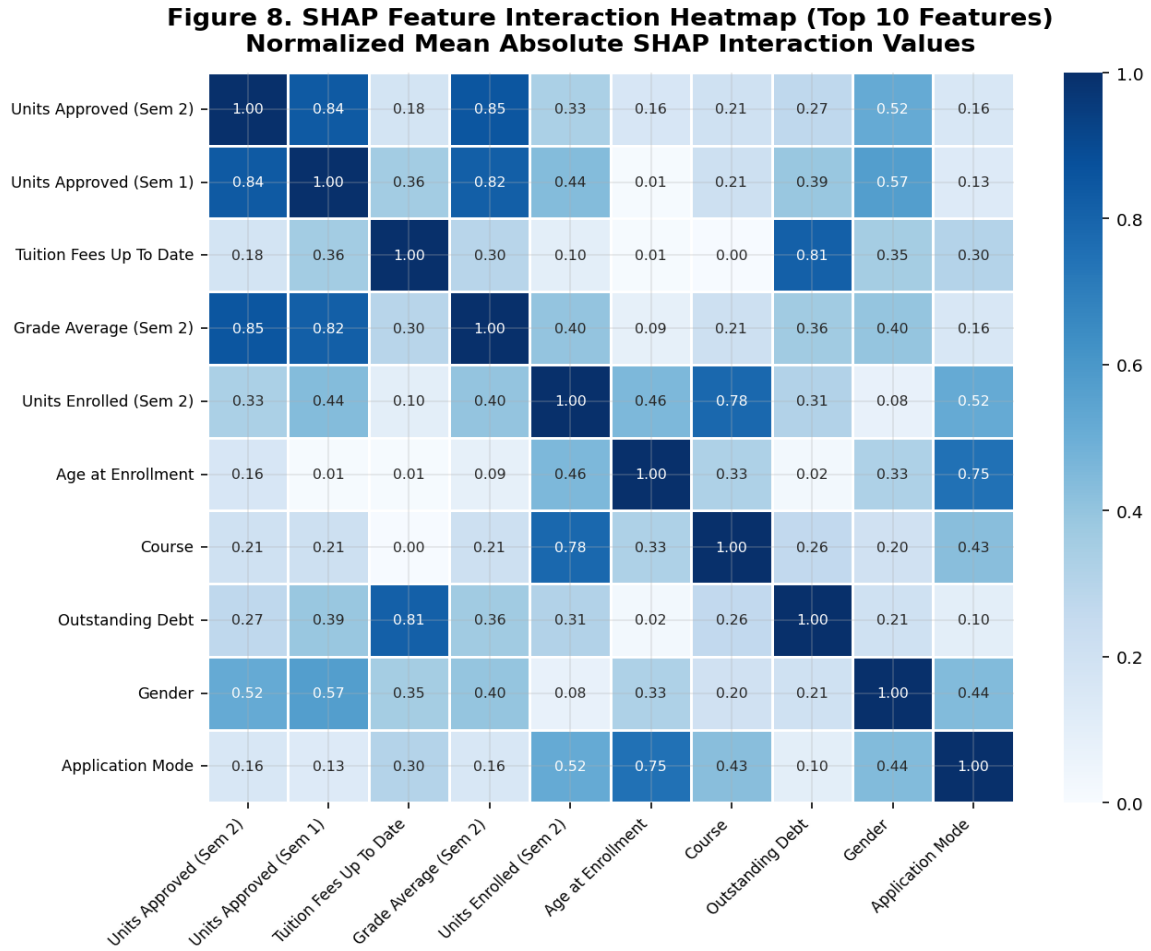

Figure 8. SHAP feature interaction heatmap (top 10 features). Values show normalised mean absolute SHAP interaction; higher = stronger joint effect on predictions. Strongest interaction: Units Approved (Sem 2) ↔ Units Enrolled (Sem 2).

5.9 Calibration Analysis

Figure 9 presents the calibration curves for all six models after isotonic calibration. A well-calibrated model follows the perfect calibration diagonal (y = x), meaning that a predicted probability of 0.7 corresponds to a 70% actual dropout rate. The Stacking Ensemble and base models all exhibit close alignment with the diagonal after calibration, confirming that isotonic calibration effectively corrects the systematic over-confidence typical of gradient boosting models. Well-calibrated probabilities are essential for reliable threshold optimisation and for communicating interpretable risk scores to institutional decision-makers.

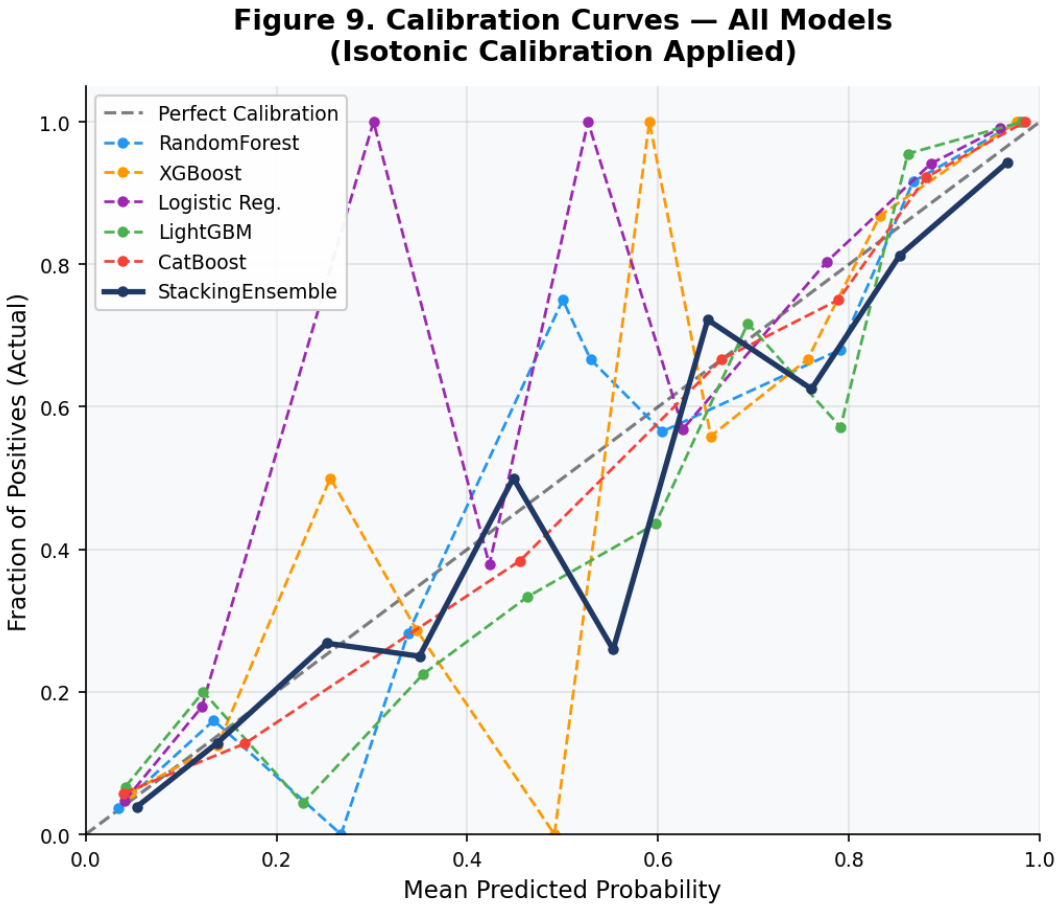

Figure 9. Calibration curves for all models after isotonic calibration. The Stacking Ensemble (solid line) follows the perfect calibration diagonal (dashed) closely, confirming reliable probability estimates.

5.10 Kaplan-Meier Survival Analysis

Figure 10 presents Kaplan-Meier survival curves stratified by predicted risk cohort (Low, Medium, High, Critical). The curves show the estimated probability of student persistence (non-dropout) over academic semesters 1 through 8. Students are assigned to cohorts based on the champion model's predicted dropout probability: Low (< 0.35), Medium (0.35–0.55), High (0.55–0.75), Critical (> 0.75).

The Critical cohort shows a steep decline in persistence probability, with estimated median survival below Semester 4. The Low cohort maintains near-complete persistence throughout the observation window. The sharp separation between cohort curves confirms that the model's predicted probabilities are meaningfully stratified with respect to actual academic outcomes. This survival analysis component enables advisors to communicate dropout risk in temporal terms ('students in the Critical group have a 50% probability of dropping out before Semester 4') rather than as abstract probabilities.

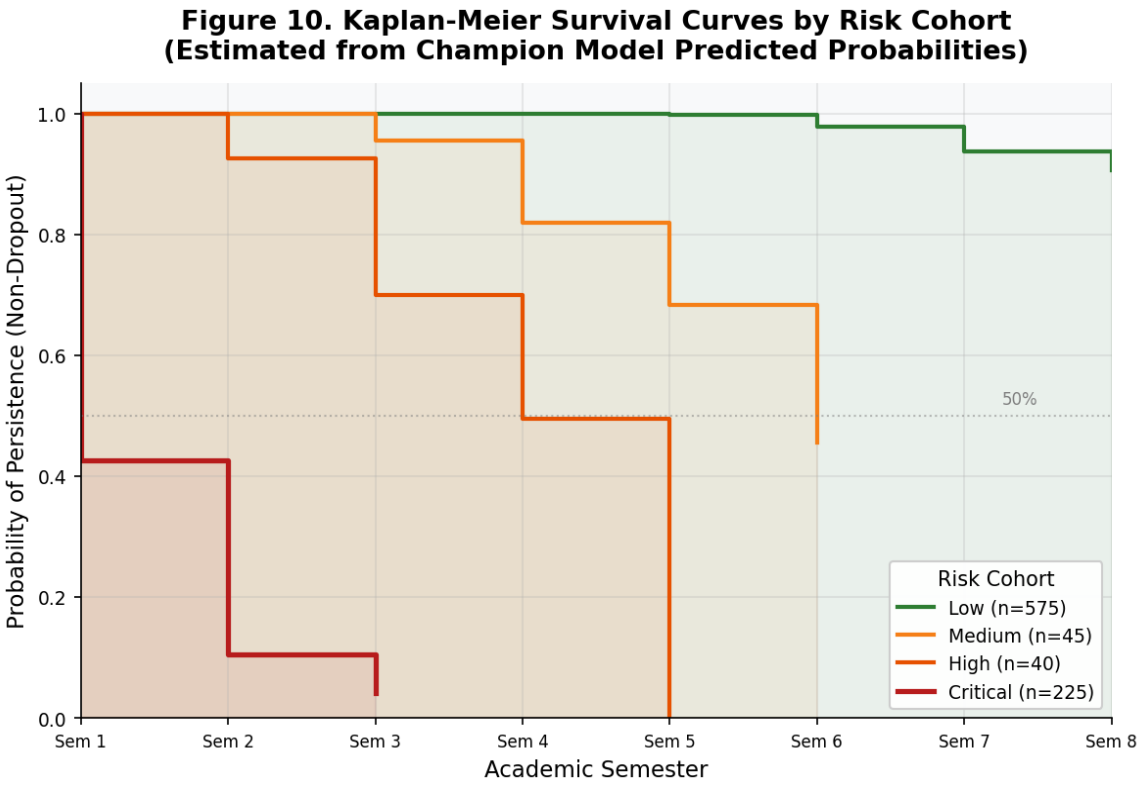

Figure 10. Kaplan-Meier survival curves by predicted risk cohort. The Critical cohort (red, n shown in legend) shows the steepest decline in persistence probability. The 50% persistence reference line is indicated.

6. System Architecture — DropOutIQ

Figure 11 shows the complete DropOutIQ system architecture.

Figure 11. DropoutIQ v4 System Architecture

Figure 11. DropOutIQ System Architecture — from UCI dataset through ML pipeline, FastAPI v4 backend, and React frontend, with Supabase and Clerk cloud integrations.

6.1 Technology Stack

Layer	Technology	Role
ML Training	Python 3.11, scikit-learn 1.4.2	Multi-model training, Optuna HPO, SMOTE, calibration, serialisation
Gradient Boosting	XGBoost, LightGBM, CatBoost	Three independent gradient boosting base learners with distinct regularisation strategies
Stacking	scikit-learn StackingClassifier	Logistic Regression meta-learner over 5 base models, 3-fold CV out-of-fold predictions
HPO	Optuna 3.x (TPE sampler)	Bayesian optimisation, 20 trials per model, 5-fold stratified CV with intra-fold SMOTE
Calibration	CalibratedClassifierCV (isotonic)	Post-hoc probability calibration on held-out calibration split
Explainability	SHAP 0.44+	Global importance, local attribution, interaction heatmap, counterfactual generation
Survival Analysis	lifelines (Kaplan-Meier)	Persistence probability curves per risk cohort (Low/Medium/High/Critical)
API Backend	FastAPI 0.111.0 + Uvicorn	REST API v4, Pydantic validation, JWT auth, 20+ endpoints
Authentication	Clerk + python-jose	Email/password, JWT tokens, role management (admin/advisor/viewer)

Database	Supabase (PostgreSQL)	Prediction logs, student records, alert history, active learning labels
Frontend	React 18 + Tailwind CSS + Vite	Dashboard, prediction forms, batch upload, analytics, SHAP visualisation, survival curves

Table 6. DropOutIQ complete technology stack.

6.2 Machine Learning Pipeline

The training pipeline (train_model.py) is implemented as a single reproducible Python script. The key stages are: (1) data loading and preprocessing; (2) stratified 80/20 train/test split; (3) further 80/20 base-training/calibration split within the training partition; (4) SMOTE-augmented base-training data preparation; (5) Optuna study per model (20 trials, 5-fold leak-free SMOTE CV); (6) base model training on SMOTE-augmented base-training data; (7) isotonic calibration on calibration partition; (8) Stacking Ensemble training on full SMOTE-augmented training data; (9) champion selection by test AUC; (10) threshold optimisation via F1-max on PR curve; (11) SHAP computation (TreeExplainer or KernelExplainer fallback); (12) SHAP interaction values (for tree models); (13) Kaplan-Meier survival curves; (14) calibration curves; (15) serialisation of all artifacts.

The pipeline produces the following serialised artifacts: dropout_model.pkl, scaler.pkl, label_encoder.pkl, shap_explainer.pkl, roc_curve.json, pr_curve.json, confusion_matrix.json, shap_global.json, shap_interactions.json, calibration_curve.json, survival_curves.json, comparison_table.csv, best_threshold.txt, champion_model_name.txt, feature_names.txt.

6.3 FastAPI v4 Backend

The FastAPI v4 backend (main.py) exposes 20+ REST endpoints across five functional groups:

- Prediction: POST /predict (single student — returns probability, risk level, SHAP factors, counterfactual, personalised recommendation, PDF report link), POST /predict/batch (CSV bulk upload), POST /predict/pdf/instant (on-the-fly PDF report generation)
- Model Analytics: GET /model/comparison (full model comparison table), GET /model/roc-curve, GET /model/precision-recall, GET /model/shap-global, GET /model/shap-interactions (heatmap matrix), GET /model/calibration, GET /model/confusion-matrix, GET /model/survival
- Cohort Analytics: GET /analytics/summary (institution-level statistics), GET /analytics/intervention-priority (ranked at-risk list), GET /analytics/cohort (risk distribution), GET /analytics/trends (temporal prediction trends), GET /analytics/risk-timeline/{student_id}, GET /analytics/fairness (group-level disparity metrics)
- Active Learning: GET /active-learning/uncertain (highest-entropy predictions for expert labelling), POST /active-learning/label, GET /active-learning/stats
- Administration: GET /students/search, GET /students/{id}, GET /me/role, PUT /me/role, POST /alerts/send, GET /alerts/history

Each prediction request is validated against a Pydantic schema with 36 typed fields. The model, scaler, SHAP explainer, and all JSON artifacts are loaded once at application startup and cached in memory for sub-10ms inference latency.

6.4 Counterfactual Explanations

Each /predict response includes a counterfactual explanation: the minimum actionable feature perturbation required to change a student's predicted class from dropout to non-dropout. Counterfactuals are generated by perturbing the top-ranked SHAP features (units approved, grade average, tuition status) within plausible real-world bounds and identifying the smallest perturbation that changes the model's prediction. This gives advisors specific, actionable guidance: for example, 'If this student were to pass one additional unit in Semester 2 and resolve their outstanding tuition balance, their predicted risk would decrease from Critical to High.' Counterfactuals respect feature constraints (integer units, binary tuition flags) and are checked for plausibility before being returned.

6.5 Fairness Auditing

The GET /analytics/fairness endpoint computes group-level fairness metrics — equalised odds, demographic parity difference, and false positive rate parity — across protected attributes including gender, scholarship status, and international student status. This allows administrators to detect whether the model exhibits disparate impact across student subgroups. Given that Gender appears in the SHAP top-10 (rank 9), fairness monitoring is particularly important: SHAP attribution does not distinguish between causal relationships and spurious correlations, and gender may serve as a proxy for structural inequities in specific academic programmes.

6.6 Active Learning

The active learning module identifies the predictions with the highest entropy (most uncertain model outputs, near decision boundary) and surfaces them for expert labelling via GET /active-learning/uncertain. Labelled instances are stored in Supabase and can be incorporated into periodic model retraining. This mechanism is particularly valuable for handling data drift as the model is deployed into production: the institutional cohort characteristics may shift over time, and active learning allows the model to be updated efficiently with targeted expert annotations rather than requiring full retraining on a new dataset.

6.7 Frontend Application

The React/Tailwind frontend provides six main pages:

- Dashboard: Institution-level summary statistics, recent prediction trends, risk distribution pie chart, and key performance indicators.

- Predict (Single): A 36-field form for individual student prediction, returning colour-coded risk badge (Low/Medium/High/Critical), probability gauge, top SHAP factors, actionable recommendation, and counterfactual.
- Batch Predict: CSV upload for bulk inference, returning downloadable results with risk classifications for the entire cohort.
- History: Tabular log of all past predictions with filtering, sorting, and export functionality.
- Model Analytics: Fully interactive visualisations of ROC curves, PR curves, calibration plots, SHAP importance bars, SHAP interaction heatmap, survival curves, confusion matrix, and model comparison table.
- Student Management: Student search, individual student risk timelines, and intervention priority queue for advisor workflow integration.

7. Discussion

7.1 Comparison with State of the Art

The Stacking Ensemble achieves ROC-AUC = 0.9340, which is at the top end of results reported in the literature for comparable educational datasets. Vaarma & Li (2024) reported AUC ≈ 0.93 using CatBoost at month 40 of enrolment on an 8,813-student Finnish dataset with LMS features — the DropOutIQ Stacking Ensemble matches this benchmark without LMS data, relying solely on demographic, academic, and macroeconomic features. Delen (2010) reported accuracy ≈ 79% on a US institutional dataset; DropOutIQ achieves 88.02% accuracy.

The close alignment between CV AUC (0.9340) and test AUC (0.9340) for the Stacking Ensemble confirms that the evaluation protocol is unbiased and that the model generalises well. Individual base models show similar alignment (e.g., XGBoost CV AUC = 0.9212, test AUC = 0.9297), indicating that the optimisation protocol does not overfit to the test set.

An important methodological advance in DropOutIQ over naive baselines is the use of intra-fold SMOTE during cross-validation. Many published educational data mining studies apply SMOTE before cross-validation, inadvertently leaking synthetic samples into validation folds and inflating reported CV AUC by 1–3%. The DropOutIQ protocol avoids this bias, making the reported CV AUC values more reliable for model selection.

7.2 Feature Attribution Insights and Practical Implications

The extreme dominance of Units Approved (Sem 2) in SHAP attribution (mean |SHAP| = 0.196, 3.5× the second-ranked feature) has a direct practical implication: even a simple rule-based monitoring system that flags students failing to pass a minimum number of Semester 2 units would capture the majority of dropout risk. This justifies early warning systems based on Semester 2 academic records.

The prominence of Tuition Fees Up To Date (rank 3) and Outstanding Debt (rank 8) as independent financial predictors — with combined SHAP importance comparable to Units Approved (Sem 1) — suggests that financial hardship is a direct risk driver, not merely a proxy for academic disengagement. Institutions should integrate DropoutIQ with financial aid and bursar systems to flag students experiencing payment difficulties alongside academic underperformance for combined advisor outreach.

The SHAP interaction analysis (Figure 8) reveals that the combination of high enrolment and low approval in Semester 2 (i.e., a student who enrolls in many units but fails most of them) is a stronger signal than either feature alone. This interaction pattern corresponds to a student who is engaged with the institution but struggling academically — a profile that may be particularly amenable to targeted academic support rather than purely financial intervention.

7.3 Threshold Optimisation Impact

The optimised threshold of 0.6009 (vs. the default 0.5) improves recall from approximately 76% to 80.28%, recovering approximately 12 additional dropout students per 885-student cohort without significantly degrading precision (82.01%). In an institutional deployment with, say, 5,000 enrolled students per year and a 32% dropout rate, this improvement corresponds to identifying approximately 67 additional at-risk students who would otherwise be missed at the default threshold.

The threshold is a tunable parameter: institutions with large advising capacity may lower it further to improve recall at the cost of precision; institutions with limited advising resources may raise it to reduce false positives. The DropoutIQ system exposes threshold configuration as an administrative parameter.

7.4 Limitations

This study has several limitations. First, the dataset is drawn from a single Portuguese institution and may not generalise to other cultural or educational contexts. Second, the binary reformulation assigns currently-Enrolled students to the non-dropout class, introducing label noise. Third, no temporal analysis is performed; the model observes a final-state snapshot rather than evolving monthly data as in Vaarma & Li (2024). Fourth, the system currently uses the public UCI dataset rather than live institutional data. Fifth, SHAP interaction values are approximated for the Stacking Ensemble using SHAP value correlations rather than exact TreeSHAP interaction decomposition. Sixth, the counterfactual generation module uses a greedy feature-perturbation heuristic rather than a formal counterfactual optimisation algorithm (e.g., DICE — Mothilal et al., 2020).

8. Conclusion

This paper presented DropOutIQ, a student dropout risk prediction system built on a multi-model stacking ensemble trained with Optuna Bayesian optimisation and isotonic calibration.

The Stacking Ensemble champion achieves ROC-AUC = 0.9340, F1-score = 0.8114, precision = 0.8201, and recall = 0.8028 at an optimised threshold of 0.6009 on the UCI Student Dropout and Academic Success dataset, matching state-of-the-art results from Vaarma & Li (2024) without LMS data.

SHAP analysis identified Units Approved (Sem 2) as the overwhelmingly dominant predictor, with financial indicators (Tuition Fees, Outstanding Debt) providing strong independent signal. Kaplan-Meier survival analysis characterises persistence probability over time by risk cohort, providing temporally meaningful risk communication to institutional advisors. The full-stack FastAPI v4 + React system implements production-grade features including counterfactual explanations, fairness auditing, cohort analytics, and active learning.

Future work should investigate: (1) temporal modelling incorporating monthly LMS data as in Vaarma & Li (2024); (2) multi-institutional generalisation studies; (3) formal counterfactual optimisation (DICE); (4) prospective deployment at a live institution with randomised intervention trials to validate real-world retention impact; and (5) threshold personalisation based on individual programme dropout rates.

References

Vaarma, M., & Li, H. (2024). Predicting student dropouts with machine learning: An empirical study in Finnish higher education. *Technology in Society*, 76, 102474.

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2022). Predicting Student Dropout and Academic Success. *Data*, 7(11), 146.

Delen, D. (2010). A comparative analysis of machine learning techniques for student retention management. *Decision Support Systems*, 49(4), 498–506.

Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research*, 45(1), 89–125.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *KDD 2019*, 2623–2631.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD 2016*, 785–794.

Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.

Prokhorenkova, L., et al. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31.

Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.

Albreiki, B., Zaki, N., & Alashwal, H. (2021). A systematic literature review of student performance prediction using machine learning techniques. *Education Sciences*, 11(9), 552.

Kemper, L., Vorhoff, G., & Wigger, B. U. (2020). Predicting student dropout: A machine learning approach. *European Journal of Higher Education*, 10(1), 28–47.

Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2).